

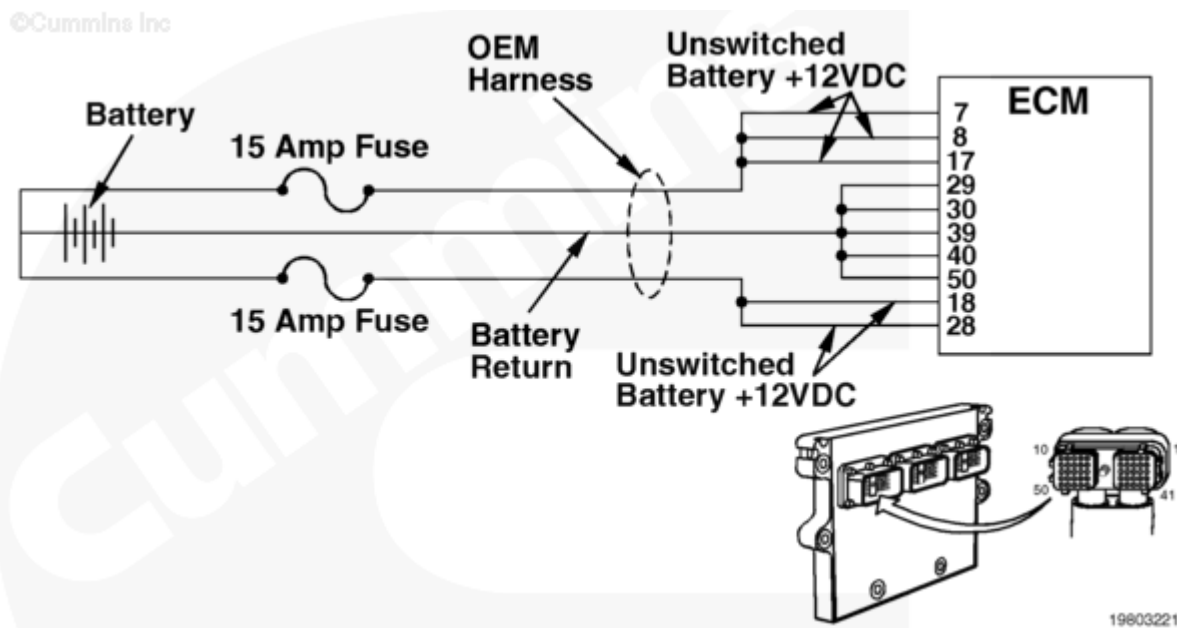
Fault Code: 597 (Integrated)

Electrical Charging System Voltage Low - Data Valid But Below Normal Operational Range - Moderately Severe Level

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 597 PID(P): P167 SPN: 67 FMI: 1/18 Lamp: Yellow SRT:	Electrical Charging System Voltage Low - Data Valid But Below Normal Operational Range - Moderately Severe Level. The ICON™ system has restarted the engine three times in 7 hours due to low battery voltage.	The engine will run continuously. The ICON™ system will not be disabled. Accessories will not be on.



Circuit Description

The engine electronic control module (ECM) receives unswitched battery input through the OEM harness. There are two in-line 15-amp fuses in the unswitched battery wire of the OEM harness to protect the engine harness from overheating. The battery return wires are connected directly to the negative (-) battery post.

Component Location

The engine ECM is connected to the battery by the OEM harness. This direct link provides a constant power supply for the engine ECM. The location of the battery will vary with the OEM. Refer to the OEM troubleshooting and repair manual for the battery location.

Shoptalk

Make certain that the engine ECM unswitched battery supply is coming directly from the battery and **not** the starter.

The following are possible causes of this fault:

- Undercharged batteries caused by a faulty alternator or regulator
- High-current devices on the vehicle such as refrigerators, citizens band radio amplifiers, numerous exterior lights, or other accessories.

Warnings and Cautions

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

CAUTION

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement:

Part Number 3822758 - male Deutsch/AMP/Metri-Pack test lead

Part Number 3822917 - female Deutsch/AMP/Metri-Pack test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Read all fault codes.	
	STEP 1A. Read the fault codes with INSITE™ electronic service tool or flash out with the ICON™ lamp.	Fault Code 597 inactive

STEPS		SPECIFICATIONS
STEP 2.	Check the equipment battery charging system.	
	STEP 2A. Inspect the battery and alternator cable connections.	Connections clean and tight
	STEP 2B. Monitor the battery voltage.	Less than 0.2 VDC
	STEP 2C. Check the charging system.	See Procedure 209-017
STEP 3.	Clear the fault codes.	
	STEP 3A. Disable the fault code.	Fault Code 597 inactive

STEP 1. Read all fault codes.

STEP 1A. Read the fault codes with INSITE™ electronic service tool or flash out with the ICON™ lamp.

Conditions :

- Turn keyswitch ON.

Action	Specification/Repair	Next Step
	Fault Code 597 inactive	3A
	Fault Code 597 active	3A

STEP 2. Check the equipment battery charging system.

STEP 2A. Inspect the battery and alternator cable connections.

Conditions :

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
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- Corrosion - Loose connections.	Connections clean and tight	2B
	Repair the damaged connections Tighten the battery and alternator connections, and clean the terminals, refer to the OEM troubleshooting and repair manual.	3A

STEP 2B. Monitor the battery voltage

Conditions : Turn keyswitch ON.		
Action	Specification/Repair	Next Step

- Connect the INSITE™ electronic service tool
- Record the battery voltage on the monitor screen
- Connect a digital volt meter to the battery pack
- Record the voltage
- Calculate the difference between the INSITE™ electronic service tool reading and the digital volt meter reading.

Less than 0.25 VDC

2C

Repair or replace the engine harness.

3A

- Refer to Procedure 019-043
(/qs3/pubsys2/xml/en/procedures/83/83-019-043.html) in Troubleshooting and Repair Manual, CELECT Plus Engines, Bulletin 3666084, or
- Procedure 019-043
(/qs3/pubsys2/xml/en/procedures/82/82-019-043-tr.html) in Troubleshooting and Repair Manual, Electronic Control System, ISM, Bulletin 3666266, or
- Procedure 019-031
(/qs3/pubsys2/xml/en/procedures/72/72-019-031.html) in Troubleshooting and Repair Manual, Electronic Control System, Signature and ISX, Bulletin 3666259, or
- Procedure 019-043
(/qs3/pubsys2/xml/en/procedures/88/88-019-043.html) in Troubleshooting and Repair Manual, Electronic Control System, CM870 ISM, Bulletin 4021381, or
- Procedure 019-043
(/qs3/pubsys2/xml/en/procedures/70/70-019-043.html) in

	Troubleshooting and Repair Manual, Electronic Control System, CM870 Signature and ISX, Bulletin 4021334, or Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/12/12-019-043.html) in Troubleshooting and Repair Manual, Electronic Control System, CM875 ISM, Bulletin 4021477.	
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STEP 2C. Check the charging system.

Conditions :

- Start the engine.

Action	Specification/Repair	Next Step
Measure the voltage output of the alternator. Note: Refer to Procedure 209-017 (/qs3/pubsys2/xml/en/procedures/97/97-209-017.html).	Refer to Procedure 209-017 (/qs3/pubsys2/xml/en/procedures/97/97-209-017.html).	3A
	Refer to the OEM to correct the problem.	3A

STEP 3. Clear the fault codes.

STEP 3A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
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• Start the engine, and let it idle. • Verify that Fault Code 597 is inactive using INSITE™ electronic service tool. • Erase the inactive fault codes using INSITE™ electronic service tool.	Fault Code 597 inactive	Repair complete
	Return to the troubleshooting steps, or contact the local Cummins Authorized Repair Location if all the steps have been completed and rechecked.	Appropriate troubleshooting charts

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